INCH-POUND

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SUPERCEDING
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MILITARY SPECIFICATION

LEAD AZIDE RD-1333

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers lead azide RD-1333 for use in the loading of detonators, blasting caps, percussion primers, fuzes, and in the manufacture of priming compositions.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: RDAR-QES-E, Picatinny Arsenal, New Jersey 07806-5000 or e-mailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.daps.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-51132	-	Sodium Carboxymethyl Cellulose (For Ammunition Use)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), first article inspection shall be performed in accordance with 4.3.

3.2 Conformance inspection. Conformance inspection shall be performed in accordance with 4.4.

3.3 Crystal growth control agent. Lead Azide RD-1333 shall be precipitated from a solution of Sodium Carboxymethyl Cellulose (NaCMC) which shall be used as the crystal growth control agent.

3.4 Physical characteristics.

3.4.1 Color. The color shall be white to buff.

3.4.2 Bulk powder. Powder shall be free flowing.

3.4.3 Particle size and shape. Particles shall be irregular in size and shape or regular bars of material (rectangles, pointed bars, etc) which may be slightly translucent. No transparent needle shaped crystals (length to width of 7 to 1 or greater, similar to those in Figure 4) or platelets shall be present (similar to those in Figure 5).

3.5 Chemical tests.

3.5.1 Purity. The purity shall be 97.5 percent minimum (min).

3.5.2 pH. The lead azide shall not be greater than a pH of 7.5 and shall not be less than a pH of 5.0.

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3.5.3 Solubility in water. The solubility of the lead azide in water shall not be greater than 1.0 percent.

3.5.4 Matter insoluble in nitric acid. The insoluble matter of the lead azide in nitric acid shall not be greater than 0.05 percent.

3.5.5 Iron and copper. There shall be no more than 3 parts per million (ppm) of iron and no more than 3 ppm of copper present in the prepared test solutions.

3.5.6 Chlorides, nitrates and acetates. There shall be no more than 4 ppm of chlorides, 15 ppm of nitrates and 72 ppm of acetates in the prepared test solutions.

3.5.7 Lead carboxymethyl cellulose. The carboxymethyl cellulose when determined as the lead salt shall be 0.50 percent min. and 1.20 percent maximum (max).

3.6 Bulk density. The bulk density shall be 1.0 gram (g) per milliLiter (mL) min.

3.7 Workmanship. The lead azide shall be free from grease, wood slivers or chips, rubber or plastic particles or any other foreign particles (see 6.8).

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4. VERIFICATION

TABLE I. Requirement/Verification cross reference matrix.

Methods of Verification		Classes of Verfication						
1 – Analysis 2 – Demonstration 3 – Examination 4 – Test		A – First Article B – Conformance						
Section 3 Requirements	Description	Verification Method				Verification Class		Section 4 Verification Method
		1	2	3	4	A	B	
3.1	First article inspection	X		X	X	X		4.3
3.2	Conformance inspection	X		X	X		X	4.4
3.3	Crystal growth control agent	X				X	X	4.6.4
3.4.1	Color			X		X	X	4.6.1.1/4.5.1
3.4.2	Bulk powder			X		X	X	4.6.1.2/4.5.1
3.4.3	Particle size and shape			X		X	X	4.6.1.3/4.5.1
3.5.1	Purity				X	X	X	4.6.2.1/4.5.1.1
3.5.2	pH				X	X	X	4.6.2.2/4.5.1.1
3.5.3	Solubility in water				X	X	X	4.6.2.3/4.5.1.1
3.5.4	Matter insoluble in nitric acid				X	X	X	4.6.2.4/4.5.1.1
3.5.5	Iron and copper				X	X	X	4.6.2.5/4.5.1.1
3.5.6	Chlorides, nitrates, and acetates				X	X	X	4.6.2.6/4.5.1.1
3.5.7	Lead carboxymethyl cellulose				X	X	X	4.6.2.7/4.5.1.1
3.6	Bulk density				X	X	X	4.6.3/4.5.1
3.7	Workmanship			X		X	X	4.6.5/4.5.1

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.3)

b. Conformance inspection (see paragraph 4.4)

4.2 Inspection conditions. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in 4.4, 4.5, and 4.6.

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4.3 First article. When specified, a sample shall be subjected to first article verification in accordance with Table II. The testing shall apply to three (3) consecutive batches of lead azide that have been produced by the contractor using the same production process, procedures, and equipment as will be used in regular production. For a continuous process, samples shall not be pulled until steady state has been reached. All material before steady state shall be discarded. All materials, except packaging, shall be obtained from the same sources as will be used in regular production.

4.3.1 First article quantity. The first article inspection shall be performed on the quantities listed in Table II.

4.3.2 Inspection to be performed. The first article inspection shall be performed in accordance with Table II.

4.3.3 First article rejection. If any item of the sample fails to comply with all first article requirements, the first article shall be rejected.

TABLE II. First article tests and inspection.

Examination or Test	No. of Consecutive Batches	Requirement Paragraph	Paragraph Reference/ Inspection Method
Crystal growth control agent	3	3.3	4.6.4
Physical characteristics	3	3.4	4.6.1
Chemical tests	3	3.5	4.6.2
Bulk density	3	3.6	4.6.3
Workmanship	3	3.7	4.6.5

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4.4 Conformance inspection.

4.4.1 Lot formation. Lot formation of lead azide shall be in accordance with the lot formation requirements of MIL-STD-1916, paragraph "Formation and identification of lots and batches".

a. A lot shall consist of one or more batches of lead azide produced by one manufacturer in accordance with the same specification or same specification revision under one set of operating conditions.

b. There shall be no intermixing of batches (see 6.12). Each batch shall be identified with a batch identification number, name of manufacturer, date of manufacture as well as dry weight of lead azide contained.

c. A new makeup of any one of the solutions (i.e., lead salt, sodium azide or sodium CMC) constitutes the formation of a new lot.

4.5 Testing. (see 6.8 for precaution)

4.5.1 Sampling. The tests depicted in 4.6.1, 4.6.3 and 4.6.5 shall be performed on an individual specimen from each batch. If any samples fail to comply with the requirements specified herein, the batch shall be rejected. The tests depicted in 4.6.2 shall be performed on a composite sample obtained by removing equal quantities of lead azide from each batch of lead azide in the lot and mixing to form a composite sample.

4.5.1.1 Preparation of composite sample. The tests depicted in 4.6.2 shall be performed on a composite sample from each lot. An equal portion of damp lead azide shall be removed from each batch in the lot, by means of a conductive plastic spoon to form a sample of approximately 25 grams (dry weight). (Note: Samples may be taken directly from the process equipment or storage container). The sample shall be blended wet until uniform. Water shall be added in case the sample appears dry. The sample, thoroughly wetted, shall be transferred to an appropriate conductive container closed and secured with adhesive tape. The container shall be labeled to show the name of the material, manufacturer, plant, lot number and quantity of material in the lot. If the sample fails to comply with the requirements specified herein, the lot shall be rejected.

4.6 Test methods and procedures. (see 6.10)

4.6.1 Physical characteristics.

4.6.1.1 Color. Determine the color of the lead azide by visual examination.

4.6.1.2 Bulk powder. Determine that the powder is free flowing by visual examination.

4.6.1.3 Particle size and shape. Spread a small portion of the wet or dry sample on a clean glass slide so that approximately an area of one square inch is covered with a thin layer of widely spaced particles. Allow wet samples to dry at room temperature. Examine under an

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optical microscope using magnifications of 50x to 200x, using a systematic approach to view a significant portion of the particles. At least 1 photomicrograph of each sample shall be retained. Particles shall be similar to those in Figures 1, 2, or 3. Particles as shown in Figure 4 or 5 are unacceptable and the batches of lead azide containing such particles shall be rejected.

4.6.2 Chemical tests.

4.6.2.1 Purity. (see 6.8.2 for precaution)

4.6.2.1.1 Solutions for purity determination.

a. Ceric Ammonium Nitrate (CAN). Dilute with deionized or distilled water 60 mL of 70-72 % perchloric acid solution to approximately 700 mL, and add this solution to 60 g of reagent grade ceric ammonium nitrate. Heat and stir the mixture until solution takes place. Depending on the source CAN solutions can appear perfectly clear or have considerable insoluble matter. Allow to cool and let stand for forty eight hours. Filter the solution through a Gooch crucible fitted with GF/F filter mats to eliminate all insoluble matter.

b. Perchloric acid solution. Dilute with deionized or distilled water 60 mL of 70-72 % perchloric acid solution to a volume of 1 liter.

c. Ferrous Perchlorate solution (FP). This may be prepared in two ways.

1. Commercially available ferrous perchlorate can be used, if the color is still green. A brown color indicates the iron has oxidized to ferric and is no longer usable. Add 30 g of the solid ferrous perchlorate into a liter beaker. Add 700 mL deionized or distilled water and 60 mL of 70-72 % perchloric acid. Stir until all solid is dissolved. Filter through a Gooch crucible fitted with GF/F filter mats and make the volume up to 1 liter. Pass nitrogen gas through the solution for two hours before using. Store solution in a closed container.

2. The second method involves making the solution from iron wire and perchloric acid. Add 6 g of iron wire into a liter beaker. Add 700 mL deionized or distilled water and 60 mL of 70-72 % perchloric acid solution. Heat beaker on a steam bath in a fume hood until the metal dissolves. There will be black residue. Filter through a Gooch crucible fitted with GF/F filter mats. Make volume up to 1 liter. Pass nitrogen gas through the solution for two hours before using.

d. Ferroin indicator solution. Commercially available ferroin indicator can be used.

4.6.2.1.2 Determination of CAN/FP ratio. Pipet 40 mL of the CAN solution into a 150 mL beaker. Add 25 mL of perchloric acid solution. Add two drops of the ferroin indicator solution. The color shall be yellow. Titrate with FP. Gradually the color lightens until the

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yellow disappears, and the solution shall be a very pale blue. Continue titration drop wise with FP while stirring until the solution turns pink. Divide the volume of CAN by the volume of FP to get the CAN/FP ratio.

4.6.2.1.3 Standardization of CAN solution. Weigh about 0.3 g of sodium oxalate into a 250 mL beaker. Add a magnetic stir bar. Pipet 40 mL CAN solution into the beaker. Add 25 mL perchloric acid solution. Add two drops ferroin indicator solution. Cover the top of the beaker with a watch glass, and stir the solution magnetically until all the sodium oxalate is dissolved. If the yellow color completely disappears, then the aliquot of sodium oxalate was too large. Try again with a smaller quantity. Titrate the contents of the beaker with FP until the solution just turns pink.

Normality of CAN N = (100xA)/ {6.701 x (40 – BxC)}

Where:
A = weight of sodium oxalate in grams
B = mL of ferrous perchlorate used
C = CAN/FP ratio

4.6.2.1.4 Determination of lead azide purity. Pipet 40 ml of CAN into a 150 mL beaker. Add 25 mL of perchloric acid solution. Add two drops of ferroin. Cover the beaker with a watch glass, and place it in an ice bath for fifteen minutes. Use Teflon-coated spatulas and forceps for handling lead azide. Perform weighings in a high humidity (about 50-60 %) weighing room. Accurately weigh an amount of approximately 0.4 g of lead azide into a tared small (1-2 mL) beaker. Using the forceps, plunge the small beaker into the 150 mL beaker. Wash the forceps tips with a spray of deionized water into the beaker. Gently stir the contents of the beaker with a glass or plastic stirring rod until all the solid has reacted. If the yellow color disappears, then the lead azide sample was too large. Repeat with a smaller amount. Titrate the solution with ferrous perchlorate while stirring until the appearance of a pink color. Calculate the purity as follows (See 6.11):

Purity of Lead Azide = {14.56 x (40 - BxC) x N}/W

Where:

B = mL of ferrous perchlorate used

C = CAN/FP ratio

N = normality of CAN solution

W = weight of lead azide sample in grams

4.6.2.2 pH. Determine the pH of approximately 1.0 g specimen of lead azide by suspending in 10 mL of water using a standardized pH meter.

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4.6.2.3 Solubility in water. Mix the wet sample thoroughly, air-dry a portion on a Buchner funnel, and transfer approximately 5 grams of the damp material to a tared, filtering crucible. Heat the crucible and contents in an oven maintained at 65-70 degrees Celsius (C) to constant weight but not more than 25 hours, cool in a desiccator, and weigh. Wash the dry lead azide with five 10 mL portions of distilled water at 0 degrees to 15 degrees C, allowing each portion to remain in contact with the lead azide for three minutes. Catch the filtrate in a clean filtering flask and retain for determination described in 4.6.2.6. Aspirate for five minutes. Dry the crucible and contents in an oven maintained at 65-70 degrees C, to constant weight but for no more than 25 hours, cool in a desiccator and weigh. Calculate the loss in weight of the lead azide to percent solubility of the sample in water, on a moisture-free basis.

4.6.2.4 Matter insoluble in nitric acid. (see 6.8.3 for precaution) Mix the wet sample thoroughly, air-dry a portion on a Buchner funnel and transfer approximately 5 g of the damp material to a tared 150 mL beaker. Heat the beaker and contents in an oven maintained at 65-70 degrees C to constant weight but for not more than 25 hours, cool in a desiccator, and weigh. Add 100 mL of dilute nitric acid (1:4) to the beaker, and warm beaker with contents on a hot water bath. (A steam bath or moderately heating hot plate may be substituted. Do not let the solution boil down to dryness.) Allow any insoluble matter present to settle and decant supernatant liquid through a tared, filtering crucible (a Gooch crucible fitted with 934AH filter mats or equivalent may be substituted), catching the filtrate in a clean filtering flask. Repeat the previous operation (adding 100 mL of dilute nitric acid, warming on a hot bath and filtering) two additional times. Transfer quantitatively the residue in the beaker to the tared filtering crucible with the aid of distilled or deionized water. Remove tared, filtering crucible from filtering flask and attach to another filtering flask (save filtrate for determination described in 4.6.2.5. Simultaneously prepare an acid blank solution using the same amount of acid used in this section.) Wash contents of crucible with five 20 mL portions of five percent sodium hydroxide, allowing each portion of the sodium hydroxide to remain in contact with the residue for 1 minute. Wash crucible and contents with three 20 mL portions of cold (0-15 °C), distilled or deionized water. Aspirate for five minutes. Dry the crucible and contents in an oven maintained at 105 degrees C plus or minus 5 degrees C for 1 hour, cool in a desiccator and weigh. Convert the gain in weight of the crucible to percent insoluble matter in nitric acid.

4.6.2.5 Iron and copper.

4.6.2.5.1 Solution for iron and copper determination. Use commercially available 1000 ppm copper and iron Atomic Absorption (AA) standards. Pipet 1 ml of each into a 100 mL volumetric flask, fill to volume with deionized water and mix thoroughly. This solution is now 10 ppm in both copper and iron. Store this solution in a plastic bottle.

4.6.2.5.2 Sample preparation. Warm the filtrate reserved from 4.6.2.4 and slowly add 40 mL of 4N sulfuric acid. Do the same to the acid blank solution from 4.6.2.4. Allow the solution to digest on a steam bath or moderately heated hot plate for 1 hr. Treat the acid blank solution the same way. Filter the sample solution through a filtering crucible (or a Gooch crucible fitted with 934AH filtering mats or equivalent) catching the filtrate in a clean filtering flask. Transfer the filtrate to a clean 250 mL beaker and evaporate the contents to fuming on a hot plate. Continue the evaporation until approximately 1 mL remains. Allow the residue to cool and add

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about 30 mL water. Filter off any further precipitate and make the filtrate up to 50 mL. The blank solution should also be reduced in volume similarly to 1 mL, then it should be transferred to a 50 mL volumetric flask and the volume made up to 30 mL with cold water. 15 mL of the 10 ppm copper and iron standard shall be pipetted into the flask for the blank, and the flask filled to volume with cold water, and mixed. The sample solution and the blank solution with added standard shall be analyzed using either atomic absorption spectroscopy or Inductively Coupled Plasma (ICP) spectroscopy. The quantities of copper and iron in the sample should be less than that of the acid blank solution with added standards.

4.6.2.6 Chloride, nitrates and acetates. Divide the filtrate obtained in 4.6.2.3 into two equal portions. Retain second portion for the acetate determination in 4.6.2.6.6. Pass hydrogen sulfide through the first portion for a few minutes and allow the beaker to stand on a steam bath to remove all the hydrogen sulfide and to coagulate the precipitate. Filter off the precipitate and make the volume of the filtrate up to 50 mL. Divide filtrate into two equal 25 mL portions which will be used for chloride and nitrate determination in 4.6.2.6.2 and 4.6.2.6.4.

4.6.2.6.1 Solution for chloride determination.

a. Nitric acid. 70 percent

b. Silver nitrate. 0.1N, dissolve 1.69 g of silver nitrate in 100 mL distilled water.

c. Standard chloride solution. Dissolve 0.085g of potassium chloride in 500 mL distilled water.

4.6.2.6.2 Chloride procedure. Transfer first portion of filtrate obtained in 4.6.2.6 to a Nessler tube. Add 1 mL concentrated nitric acid and 1 mL 0.1N silver nitrate solution. Compare the turbidity with that of a standard, equivalent to 0.02 percent lead chloride, in a second Nessler tube. Place 1 mL of the standard chloride solution in a Nessler tube, add 1 mL of 0.1N silver nitrate and make up to 50 mL. The turbidity present in the chloride determination should be the same or less than that of the standard.

4.6.2.6.3 Solution for nitrate determination.

a. Diphenylamine Solution. Dissolve 0.5 g of diphenylamine in 50 mL of 95-98% sulfuric acid.

b. Standard nitrate solution. Dissolve 0.125 g lead nitrate in 500 mL of distilled water.

4.6.2.6.4 Nitrate procedure. Transfer second portion of filtrate obtained in 4.6.2.6 to a 100 ml beaker and evaporate on a steam bath to about 2 mL. Transfer, quantitatively, to a test tube, add 2 mL concentrated sulfuric acid and 1 drop diphenylamine solution. The color obtained shall be equal or less than the color obtained with that of a standard (i.e., pipet 2 mL of standard nitrate solution into a test tube, add 2 mL concentrated sulfuric acid and 1 drop of diphenylamine solution).

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4.6.2.6.5 Solutions for acetate determination.

a. Iodine solution A. Dissolve 1.27 g of iodine and 1.6 g of potassium iodide in 100 mL of distilled water.

b. Iodine solution B. Dissolve 0.127 g of iodine and 0.16 g of potassium iodide in 100 mL distilled water.

c. Lanthanum nitrate solution. Dissolve 5 g of lanthanum nitrate in 100 mL of distilled water.

d. Standard acetate solution. Dissolve 0.05 g of sodium acetate in 500 mL of distilled water.

4.6.2.6.6 Acetate procedure. Add 1 mL of iodine solution A to the filtrate obtained in 4.6.2.6 and evaporate the solution to about 2 mL on a steam bath. Cool the solution and quantitatively transfer it into a graduated test tube which has a one mL marking. Continue the evaporation in the test tube until the volume has been reduced to 1 mL. Cool and add 3 drops of iodine solution B, 3 drops of 5 percent lanthanum nitrate and 4 drops of 0.5 N ammonium hydroxide in that order, shaking after each addition. Bring the solution to a boil, remove from heat and allow it to stand for 10 minutes. Perform a comparative test starting with 25 mL of standard sodium acetate solution using the procedure of adding iodine solution A, evaporation, adding iodine solution B, evaporation, etc. The blue color produced by the test should be no greater than that produced by the standard. Any color other than blue shall require a repeat of the test procedure.

4.6.2.7 Lead carboxymethyl cellulose. (see 6.8.5 for precaution) Transfer 2.00 g of a dried sample to a 250 mL beaker. Add 75 mL of glacial acetic acid to the beaker and place beaker with contents on a steam bath, or moderately heated hot plate (do not let solution boil down to dryness). Swirl gently for a few minutes. Allow the beaker to stand on the steam bath covered with a watch glass for 30 minutes. While hot, filter through a tared sintered glass crucible or a Gooch crucible fitted with 934AH filter mats or equivalent. Use an additional 50 mL of hot glacial acetic acid to rinse beaker and crucible. Treat residue in crucible with three 20 mL portions of cold 10 percent lead acetate solution, by filling the crucible with the lead acetate solution and applying gentle suction. Wash thoroughly with hot (40-90 degrees C) distilled water, then dry at 105 plus or minus 5 degrees C for 1 hour, cool in a desiccator and weigh. Calculate the percentage of lead carboxymethyl cellulose as follows:

Percent lead carboxymethyl cellulose = (A-B) x 100/C

Where:

A = weight of crucible plus residue

B = weight of crucible

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C = dry weight of sample

4.6.3 Bulk density.

4.6.3.1 Preparation of specimen.

4.6.3.1.1 Applicable to dry lead azide. Transfer a portion of approximately 4 g to a drying dish. Place the dish and contents into an oven, maintained no higher than 70 degrees C, until it reaches a constant weight. Then cool in a dessicator to room temperature.

4.6.3.1.2 Applicable to wet lead azide. Transfer a portion of approximately 4 g to a 50 mL beaker. Decant off the excess water and add 20 to 30 mL of 95% ethyl alcohol. Gently swirl the beaker and contents for a few minutes and then decant off the alcohol. Repeat the procedure of adding alcohol, swirling and decanting once more. Transfer the contents of the beaker into a weighing dish. Place the dish and contents into an oven, maintained no higher than 70 degrees C, until it reaches a constant weight. Then cool in a dessicator to room temperature.

4.6.3.2 Charge plate method.

4.6.3.2.1 Aluminum charge plate. The aluminum charge plate shall be 3 inches long and 2 inches wide with a thickness of 0.278 inch. A single frustrum-shaped cavity near the center of the plate will have a volume of about 0.1467 mL. The cavity shall have a top diameter of 0.230 inch, a bottom diameter of 0.177 inch, and cut through the entire thickness of the plate.

4.6.3.2.2 Determine volume of charge plate cavity. Tare a weighing pan. Place charge plate on a piece of glazed paper. Fill the cavity with mercury until level with the surface of the charge plate. Remove charge plate, allowing the mercury to flow onto the glazed paper. Transfer the mercury to the pan and weigh. Determine the volume (V) of the cavity by dividing weight of the mercury used to fill the cavity by the density of mercury at the working temperature.

4.6.3.2.3 Aluminum charge plate procedure. (see 6.8.6 for precaution) Tare a weighing pan. * Place the aluminum charge plate on a piece of glazed paper. Use a plastic conductive spoon to fill the cup. Level off the excess lead azide, with the aid of a cardboard spatula, so that it is even with the surface of the plate. Take care that the excess lead azide remains on the plate. Carefully lift the plate from the glazed paper, which will allow only the lead azide in the cavity to transfer to the glazed paper. The sample on the paper is transferred to the weighing pan. Repeat the process from the * until 5 samples are collected in the weighing pan. Weigh the pan and calculate the bulk density with the following formula:

Bulk Density, g/mL = A/(5xV)

Where

A = weight of the 5 samples

V = the volume of the charge plate cavity

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4.6.3.3 Cup method.

4.6.3.3.1 Aluminum measuring cup. The aluminum cup shall be approximately 0.8 centimeter (cm) in height with an internal diameter of approximately 0.8 cm. The internal surfaces of the cup must be smooth. A handle may be affixed to the side of the cup.

4.6.3.3.2 Determine the volume of the cup. Tare the cup. Fill the cup with water, ethanol, ethanol/water mixtures, or other non-toxic fluid until level with the brim. Weigh the full cup. Calculate the volume by dividing the weight by the density of the fluid at the working temperature. Repeat procedure five times and calculate the average volume (V_C).

4.6.3.3.3 Measuring cup procedure. (See 6.8.7) Tare a weighing pan. * Place cup on a square of aluminum foil or weighing paper. Use a plastic conductive spoon to fill the cup. Level the lead azide with the brim of the cup using the spoon, allowing any excess to fall to the aluminum foil/weighing paper. Dump the contents of the cup into a tared weighing pan. Discard any lead azide on the aluminum foil/weighing paper. Repeat the procedure from the * until five cupfuls are collected in the weighing pan. Weigh the pan and determine the bulk density by the following formula:

Bulk density, g/mL = $A/(5xV_C)$

Where

A = weight of the 5 cupfuls

V_C = The calculated average volume of the measuring cup

4.6.4 Crystal growth control agent. The Sodium Carboxymethyl cellulose (NaCMC) shall be in accordance with MIL-S-51132 or approved alternate.

4.6.5 Workmanship. Visually verify that the lead azide contains no grease, wood slivers or chips, rubber or plastic particles, or any foreign particles (visible to the unaided eye).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. RD-1333 is intended for use in the loading of detonators, blasting caps, percussion primers, fuzes, and in the manufacture of priming compositions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples.
c. Requirements for Ammunition Lot Numbering, if applicable.
d. Requirement for Acceptance Inspection Equipment (AIE) designs (see 6.3)
e. Provisions for submission of acceptance inspection results for each lot of ammunition presented to the Government.
f. Certificate of Analysis and photo micrograph for each lot or shipment of product.
g. Packaging requirements.

6.3 Submission of inspection equipment designs for approval. Submit equipment designs as required to Commander, U.S. Army ARDEC, ATTN: RDAR-QEM-G, Picatinny Arsenal, NJ 07806-5000. Request letter of submittal should state the contractor, contract number, specification number, item nomenclature and classification of defects or tests. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Packing. Lead azide should be packaged and shipped in a wet condition as directed in the Code of Federal Regulations CFR 49, for manufacturers within the United States. For foreign manufacturers, all transportation regulations should be in accordance with the Foreign National Transportation Authority and the Recommendations on the Transport of Dangerous Goods, as published in document ST/SG/AC.10/1, current revisions, prepared by the U.N. Committee of Experts on the Transport of Dangerous Goods. A mixture of water plus denatured ethyl alcohol in such proportions to prevent freezing should always be used unless otherwise directed (see 6.9).

6.4.1 Level A. Up to 25 lbs. (dry weight) of lead azide should be packed wet (per 6.9) directly into a cambric diaper or similar cloth bag. A cap of the same diameter and fabric as the bag should be placed inside the bag and over the lead azide. The bag should then be tied closed and placed inside a bag made of conductive velostat. This velostat bag should then be filled with water/alcohol mixture which meets the requirements of 6.9. This water/alcohol mixture must completely cover the bag contents. The velostat bag is then tied closed and placed inside a 4 oz. or heavier duck bag. A cap of the same materials and of the same diameter as the duck bag should be placed over the velostat bag. The duck bag is then securely tied closed. This bag is now the unit bag which should be marked in accordance with 6.5. An appropriate number of bags (whose combined total dry weight of lead azide should not exceed 150 lbs.) is placed into a rubber cloth bag. Inside this bag and over the unit of bags of lead azide should be placed a cap of the same fabric and same diameter as this bag. This rubber cloth bag should then be filled with a

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water/alcohol mixture meeting requirements of 6.9 so that the unit bags inside are completely covered. The bag is then securely tied closed. This rubber bag is placed inside a strong grain bag which must be securely tied closed. The grain bag and its contents must be packed in the center of an open head steel drum conforming to UN 1A2, standards for Steel Drums, which has been lined with a heavy, close fitting jute bag. The grain bag should be entirely surrounded by not less then 3 inches of well-packed sawdust saturated by the previously mentioned water/alcohol mixture. The jute bag should then be closed by secure sewing to prevent the escape of the sawdust. The dry weight of lead azide in one outside container should not exceed 150 lbs.

6.4.2 Level B. Level B should be the same as Level A.

6.4.3 Level C. Level C should be the same as Level A.

6.5 Marking. Marking should be in accordance with MIL-STD-129.

6.5.1 Unit packaging. Each unit bag as defined in 6.4.1 should be marked in contrasting color to the bag with letters not less than ¼ inch or greater than ½ inch high with water/alcohol-proof ink as follows:

a. NSN
b. Lead Azide RD-1333
c. Quantity (dry weight) in pounds
d. Lot number
e. Batch numbers

6.5.1.1 Alternative. As an alternative, each unit bag may be identified by means of a waterproof tag imprinted with a water/alcohol-proof ink with the above information and attached with a non-metallic tie.

6.5.2 Exterior container. The side and top of each drum or barrel should be marked: “Lead Azide, Wetted, UN 0129 – DANGEROUS – DO NOT STORE OR LOAD WITH ANY HIGH EXPLOSIVE” in ½ inch high letters. The following additional information will be marked on the side of each drum or barrel in letters not less than ¼ inch or greater than ½ inch high:

a. NSN
b. Lead Azide RD-1333
c. Quantity (dry weight) in pounds
d. Lot number
e. Gross weight
f. MIL-DTL-46225

6.6 Performance oriented packaging. Prior to shipment, the contractor should ensure either that the container has been tested by either a U.S. Government approved Performance Oriented Packaging (POP) test laboratory or by the government itself in compliance with POP requirements in accordance with Title 49, Code of Federal Regulations. Test will be to a weight

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at least 10% greater than the actual gross weight or the certification weight specified on the packaging drawings, (whichever is higher) which should be marked on the tested container. POP marking, however, should not be applied to the container until verified by the U.S. Government as follows:

a. The POP test report should be prepared by the POP test laboratory in accordance with DI-PACK-81059 (Data Item Description) following the test.

b. The report must be kept on file by the contractor and must also be submitted to the U.S. Army (ARDEC Packaging Div) for verification as directed in the contract data requirements list (CDRL) DD Form 1423.

c. The US Army will review the report, prepare an endorsement letter, and submit the package to the Defense Logistics Agency (DLA).

For multi-year contracts, the contractor should re-perform POP testing at a certified POP test laboratory if:

a. The initial POP test report expires before the end of the contract (single pack POP certification is valid for a period of 1 year; combination pack POP certification is valid for a period of 2 years) or

b. There is a change in container manufacturer, design, or manufacturing processes of the exterior shipping container.

No re-test is needed if all packaging is purchased while under an unexpired POP test.

EXCEPTION TO POP MARKINGS: If the container is manufactured outside the USA, the contractor should not apply the UN POP certification mark provided in this contract (if applicable). The contractor/container manufacturer (outside the USA) is responsible to perform the UN POP certification tests and apply the marking authorized by the Transportation Competent Authority of the country of manufacture. The contractor should also provide a copy of the POP Test Report to the U.S. Army (ARDEC Packaging Div) for concurrence prior to shipment.

6.7 First article rejection. The procuring agency reserves the right to terminate inspection upon any failure of a test specimen to comply with any of the requirements.

6.8 Necessary precautions. Lead azide RD-1333 is classified as a primary explosive. All necessary precaution in the manufacture, handling, and storage of such material should be observed. During manufacture, care must be exercised to assure that the reactor is thoroughly cleaned before manufacture of each new batch. [Note: the cleaning may be mechanical (i.e., scouring with a wet sponge or a strong stream of water) or chemical (i.e., destroying the azides chemically, followed by thorough rinsing with water). However, the reactor must be cleaned chemically before preparation of every fourth batch if mechanical cleaning is chosen.] The solution of sodium carboxymethyl cellulose must be filtered prior to use in such a manner to

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preclude the presence of fibrous material. The lead solution must be filtered until clear (i.e., no suspension of lead carbonate) and must be stored in such a manner as to limit the formation of lead carbonate due to atmospheric carbon dioxide contamination.

6.8.1 Waste lead azide. To safely dispose of small quantities of waste lead azide, add all azide residues or contamination into 50 times its weight of 15 percent ceric ammonium nitrate solution.

6.8.2 Purity precaution. Caution must be exercised. Evolution of hydrazoic acid (which is toxic but odorless) occurs. Therefore, all operations where the lead azide is dissolved in an acid should be performed in a fume hood.

6.8.3 Matter insoluble in nitric acid precaution. Caution must be exercised. In this test the lead azide is dissolved in dilute nitric acid, a process which gives rise to the evolution of hydrazoic acid gas which is toxic but odorless. Therefore, this operation should be done in a fume hood.

6.8.4 Chloride, nitrates and acetates precaution. Caution must be exercised. All operations should be performed in a fume hood.

6.8.5 Lead carboxymethyl cellulose precaution. Caution must be exercised. In this test the lead azide is dissolved in glacial acetic acid, a process which gives rise to the evolution of hydrazoic acid gas which is toxic but odorless. Therefore, this operation should be done in a fume hood.

6.8.6 Aluminum charge plate procedure precaution. Perform this test behind a properly barricaded and grounded area and limit the quantity of lead azide in the area to only that which is going to be tested.

6.8.7 Measuring cup procedure. Perform this test behind a properly barricaded and grounded area and limit the quantity of lead azide in the area to only that which is going to be tested.

6.9 Water/alcohol. Due to danger of freezing conditions during shipment and storage, this material should be kept “wet’ with a 50/50 solution of water/denatured ethanol.

6.10 Test deviations. Any deviations from the test and/or sampling procedures specified herein require prior approval by the procuring activity.

6.11 Purity method validation. It is recommended to test the method first on a sample of commercially prepared high purity sodium azide. Substitute 6.5 for 14.56 in the formula as specified in 4.6.2.1.4. A value of 100 ± 0.5 % recovery inspires confidence in the values used for normality and CAN/FP ratio. Weight (W) is the weight of the sodium azide sample in grams for the validation test.

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6.12 Batch definition. A batch is defined as the lead azide manufactured under a continuous set of conditions without process interruption or change.

6.13 Photomicrographic examination. Photomicrographic examination should be conducted in accordance with 4.6.1.3 not more than 90 days prior to loading into an item.

6.14 Subject term (key word) listing.

Primary explosive
Detonators
Blasting caps
Fuzes

6.15 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes

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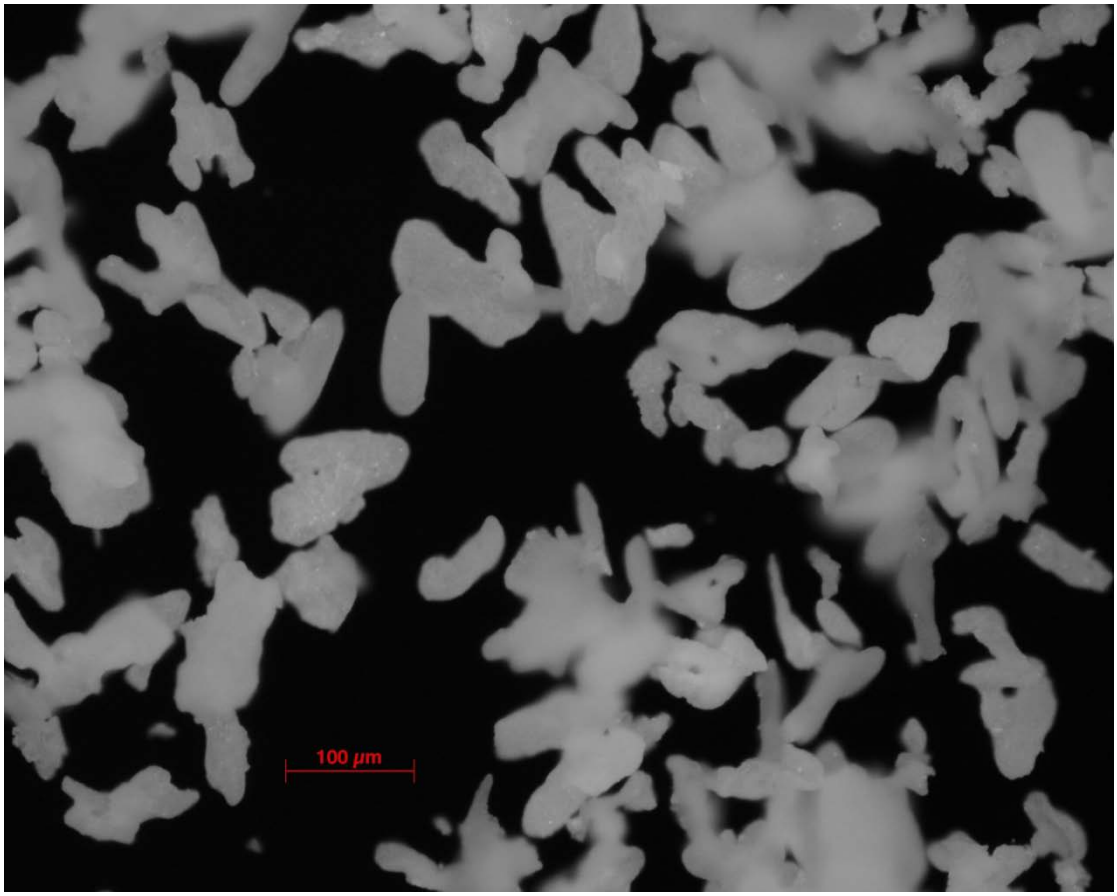


Figure 1. Acceptable RD-1333 100x.

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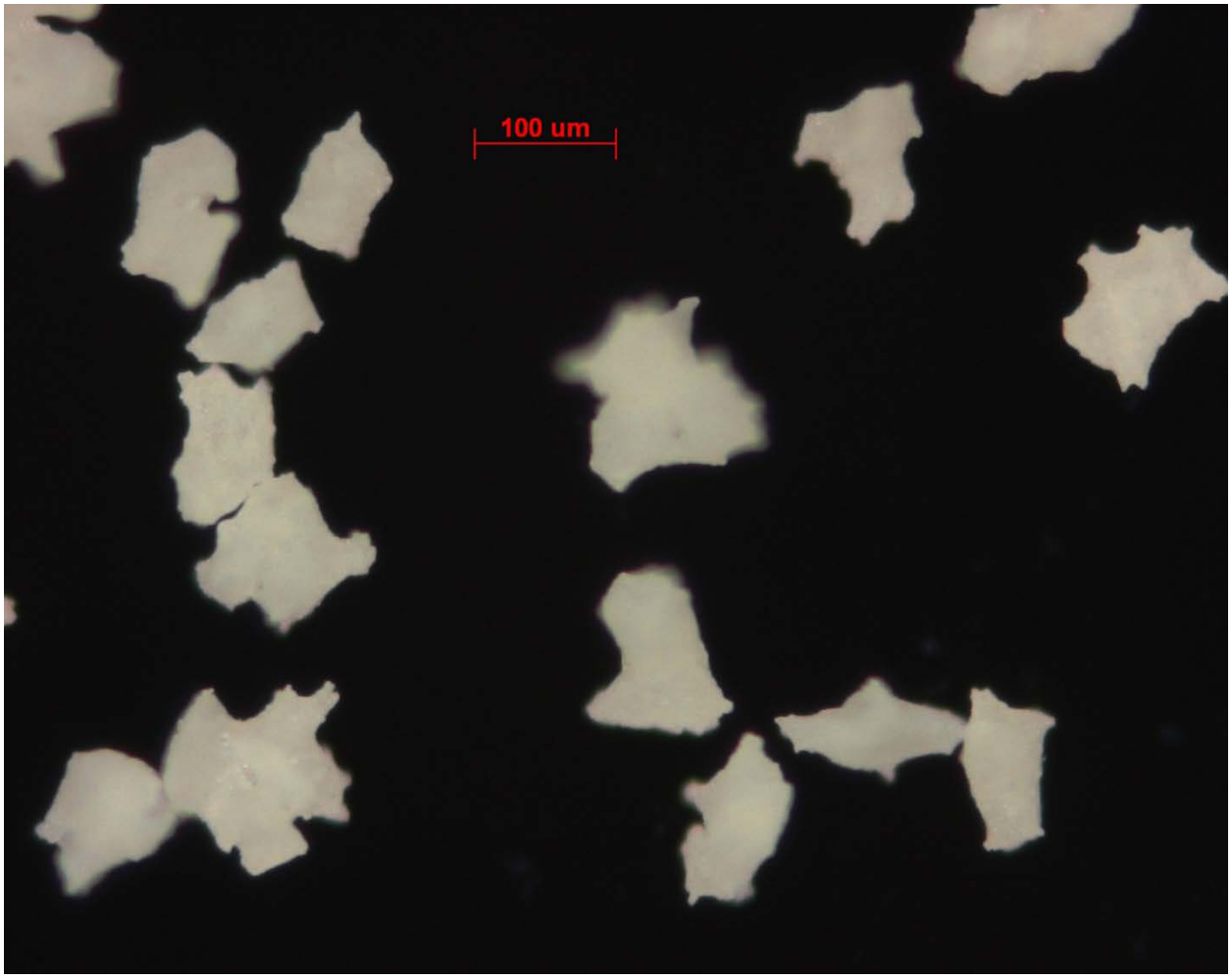


Figure 2. Acceptable RD-1333 100x.

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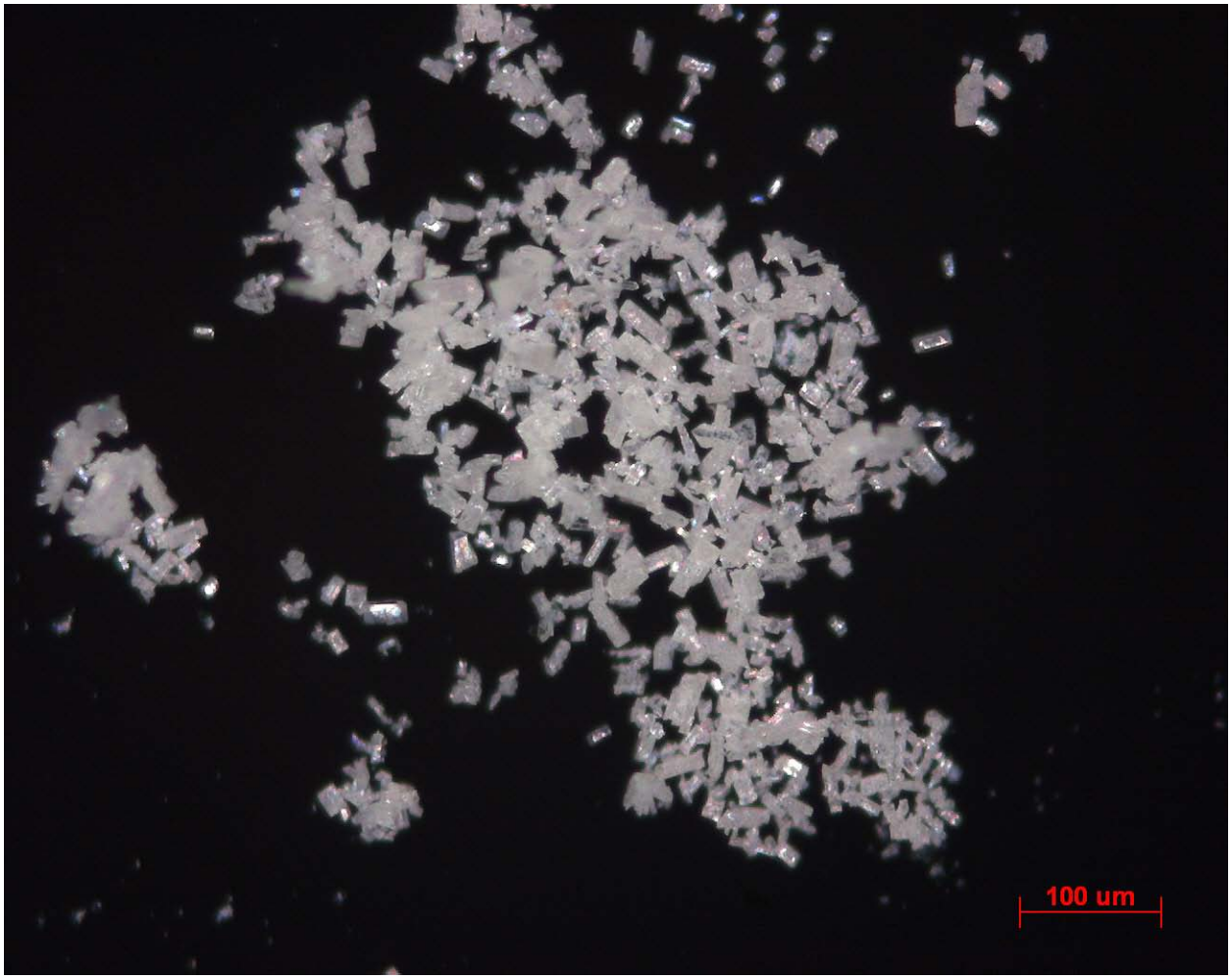


Figure 3. Acceptable RD-1333 100x.

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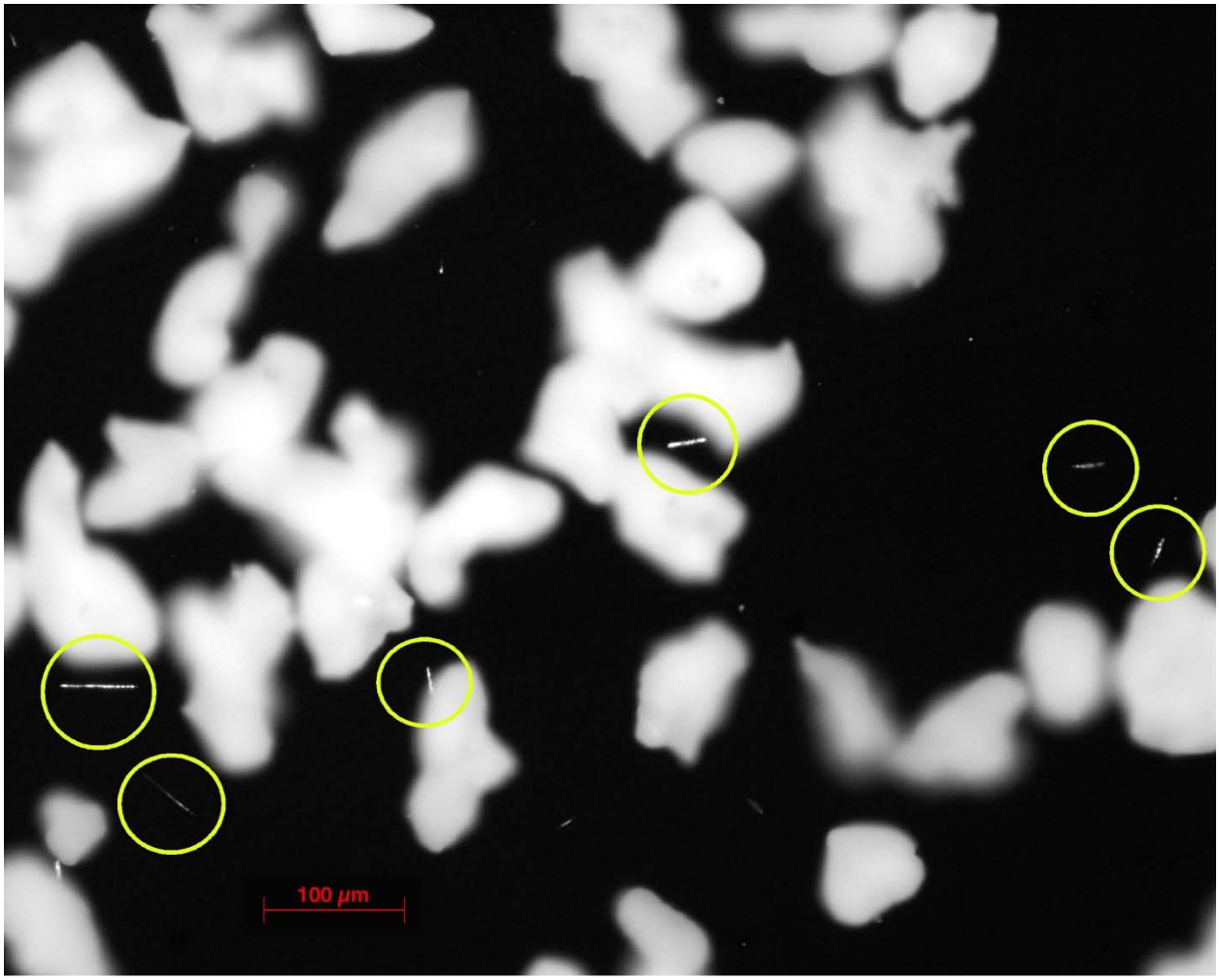


Figure 4. Unacceptable needles 100x.

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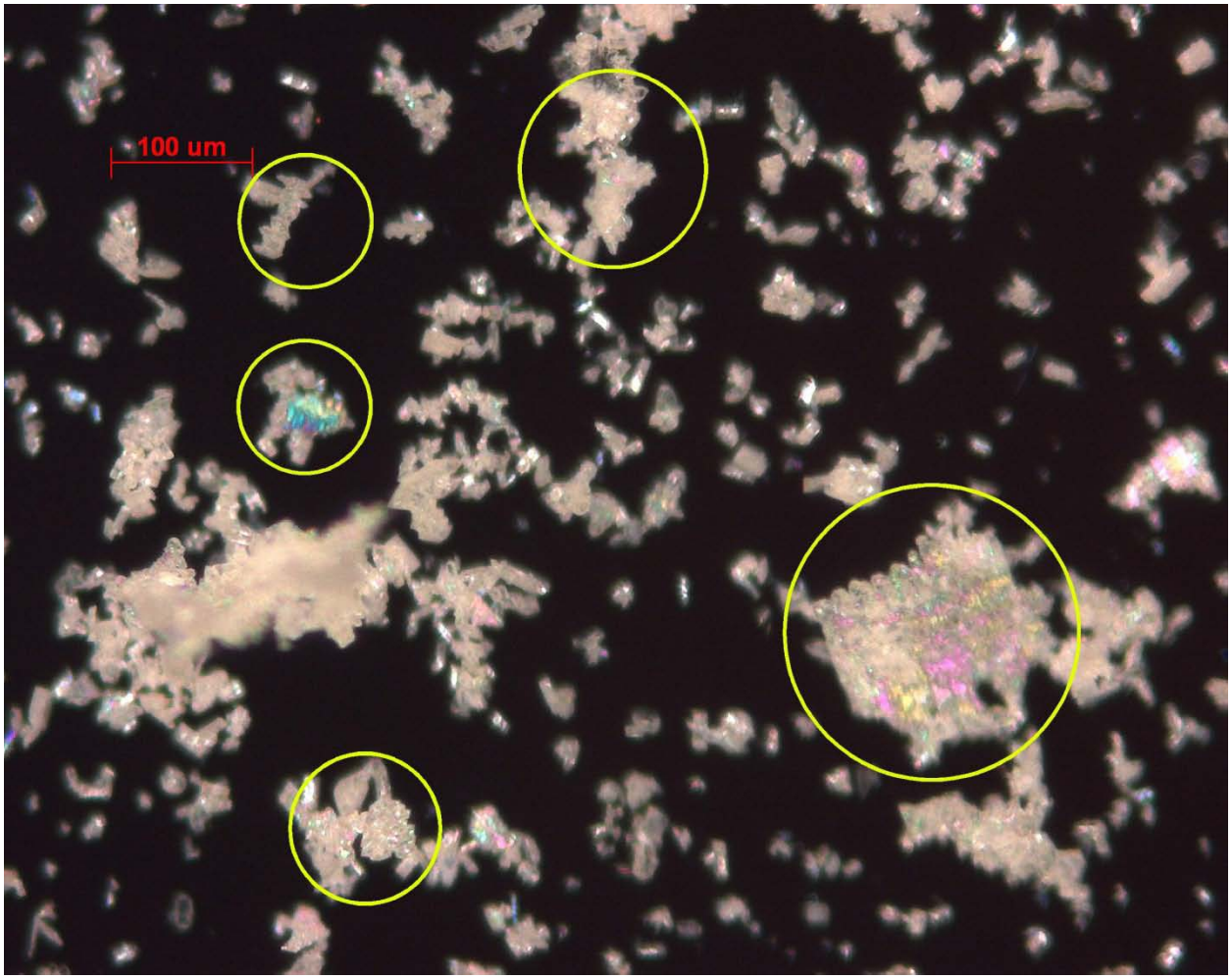


Figure 5. Unacceptable platelets 100x.

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Custodian:
Army – AR
Air Force - 11
Navy – OS

Preparing activity:
Army – AR
(Project: 1376-2010-010)

Review activities:
Army – AS
Navy – NP
Air Force – 70, 99

Civil agency:
GSA -FAS

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